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CVE5101-2 Theory-HW

1. Encrypt "CHASE YOUR DREAMS"

1(a) Caesar Cipher

∴ Key is 3, $C = (P+3) \bmod 26$

writing the alphabets

Plain: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

Cipher:

Encrypt each letter

Plaintext: CHASE YOUR DREAMS

Plain	shift +3	Cipher
C	C → D → E → F	F
H	H → I → J → K	K
A	A → B → C → D	D
S	S → T → U → V	V
E	E → F → G → H	H
Y	Y → Z → A → B	B
O	O → P → Q → R	R
U	U → V → W → X	X
R	R → S → T → U	U
D	D → E → F → G	G
R	R → S → T → U	U
E	E → F → G → H	H

A	A → B → C → D	D
M	M → N → O → P	P
S	S → T → U → V	V

Putting together at last

CHASE YOUR DREAMS — FKDVH BRXU GUNDPV

1(b) Shift Cipher with Key=7, $C = (P+7) \bmod 26$

Write the alphabet

Plaintext: ABCDEF GHIJ KLMN OPQR STUV WXYZ

Encrypt "CHASE"

C → J, H → O, A → H, S → Z, E → L = JOHZZL

Encrypt "YOUR"

Y → F, O → V, U → B, R → Y = FVBV

Encrypt "DREAMS"

D → K, R → Y, E → L, A → H, M → T, S → Z = KYLHTZ

Final Answer

CHASE YOUR DREAMS → JOHZZL FVBV KYLHTZ

1(c) ROT13 $\rightarrow$ Rotate by 13 positions

$$C = (P + 13) \bmod 26$$

Writing the alphabet

Plain text: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

1) Encrypt "CHASE"

C $\rightarrow$ P, H $\rightarrow$ U, A $\rightarrow$ N, S $\rightarrow$ F, E $\rightarrow$ R = PUNFR

2) Encrypt "YOUR"

Y $\rightarrow$ L, O $\rightarrow$ B, U $\rightarrow$ H, R $\rightarrow$ E = LBHE

3) Encrypt "DREAMS"

D $\rightarrow$ Q, R $\rightarrow$ E, E $\rightarrow$ R, A $\rightarrow$ N, M $\rightarrow$ Z, S $\rightarrow$ F = QERNZF

Final Answer

CHASE YOUR DREAMS = PUNFR LBHE QERNZF

2 Decryption

a) Caesar Cipher

WRWDOOB VHFUHW PHVVOJH

for Caesar Cipher

Decryption Formula

$$P = (C - 3) \bmod 26$$

For example, w has position 22

$$P = (22 - 3) \bmod 26$$

$$= 19$$

Position 19 is T.

$w \rightarrow T$.

So, similarly.

Plain Text: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

WRWD00B

$W \rightarrow T, R \rightarrow O, W \rightarrow T, D \rightarrow A, O \rightarrow L, O \rightarrow L, B \rightarrow Y = \text{TOTALLY}$

VHFUHW

$V \rightarrow S, H \rightarrow E, F \rightarrow C, U \rightarrow R, H \rightarrow E, W \rightarrow T = \text{SECRET}$

~~H~~

PHVVDTH

$P \rightarrow M, H \rightarrow E, V \rightarrow S, V \rightarrow S, D \rightarrow A, T \rightarrow G, H \rightarrow E = \text{MESSAGE}$

Final Answer:

TOTALLY SECRET MESSAGE

b) Shift Cipher with key = 5

M T Q A T B T W G I

$$k = 5$$

Decryption Formula

$$P = (C - 5) \bmod 26$$

M has position 12

$$P = (12 - 5) \bmod 26 = 7$$

Position 7 is H., $M \rightarrow H$

so, similarly

Plaintext: A B C D E F G H I J K L M N O P Q R S T U V X Y Z

M J Q Q T

$M \rightarrow H, J \rightarrow E, Q \rightarrow L, Q \rightarrow L, T \rightarrow O = \text{HELLO}$

B T W Q I

$B \rightarrow W, T \rightarrow O, W \rightarrow R, Q \rightarrow L, I \rightarrow D = \text{WORLD}$

Decrypt S T B D T W P Y N R J

$$P = (C - 5) \bmod 26$$

1) S T B

$S \rightarrow N, T \rightarrow E, B \rightarrow W = \text{NEW}$

2) D T W P

$D \rightarrow Y, T \rightarrow O, W \rightarrow R, P \rightarrow K = \text{YORK}$

3) Y N R J

$Y \rightarrow T, N \rightarrow I, R \rightarrow M, J \rightarrow E = \text{TIME}$

FINAL Answer:

NEW YORK TIME

c) ROT13

FRAQ ZBER PBSSRR

 $k=13$

Decryption Formula

$$P = (C - 13) \bmod 26$$

For position 5

$$P = (5 - 13) \bmod 26$$

$$= -8 \bmod 26 = 18$$

Position 18 is S

 $F \rightarrow S$

So, similarly

FRAQ = SEND

 $F \rightarrow S$ $R \rightarrow E$ $A \rightarrow N$ $Q \rightarrow D$

PBSSRR

 $P \rightarrow C, B \rightarrow O, S \rightarrow F, S \rightarrow F, R \rightarrow E, R \rightarrow E = COFFEE$

Final Answer

SEND MORE COFFEE

- 1(a) Caesar Cipher, $k=3 \rightarrow$ FKDVH BRXU GUHDPV
- 1(b) Shift Cipher, $k=7 \rightarrow$ JOHZLFVBYKYLHTZ
- 1(c) ROT13, $k=13 \rightarrow$ PUNFR LBHE QERNZF
- 2(a) Caesar Cipher, $k=3 \rightarrow$ TOTALLY SECRET MESSAGE
- 2(b) Shift Cipher, $k=5 \rightarrow$ HELLOWORLD
- 2(b) Shift Cipher, $k=7 \rightarrow$ NEW YORK TIME
- 2(c) ROT13, $k=13 \rightarrow$ SEND MORE COFFEE

formula to remember

Encryption:

$$C = (p + k) \bmod 26$$

Decryption

$$P = (c - k) \bmod 26$$

Caesar Cipher

$$k=3$$

Shift Cipher

$k = \text{given key}$

ROT13 $k=13$